

RTutor.ai Quick-Start Guide

Quantitative Summaries, Density Plots, and the Middle 95%
Rayhan Driver and V.N. Vimal Rao

0) Setup

1. Upload the Exam Grades dataset in RTutor.ai under **(1. Data) -> Browse**.
2. Confirm four columns: *Student ID*, *Year In School*, *Practice Exam Grade*, *Actual Exam Grade*.
3. Under **(2. Data Types) -> Edit**, set Student ID = Character, Year In School = Category, both grade columns = Integer (or Numeric).

1) Summary Statistics

Goal: Generate summary statistics for Practice Exam Grade.

Your prompt to RTutor.ai must:

1. Reference the Exam Grades dataset and the variable Practice Exam Grade.
2. Request these statistics in order: count, mean, standard deviation, min, Q1, median, Q3, max.
3. Ask for rounding to two decimal places and a consistent display order.

Then add the middle 95%:

- In a follow-up prompt, ask RTutor.ai to compute the 2.5th and 97.5th percentiles (quantiles 0.025 and 0.975) and label them as the middle 95% range.

Check: Does your output show all eight statistics plus both percentile cutoffs? If RTutor.ai is confused, try using the word “quantiles” with decimal cutoffs (0.025, 0.975).

2) Density Plot

Goal: Create a density plot of Practice Exam Grade.

Your prompt must include:

1. Specify the Exam Grades dataset and the variable Practice Exam Grade.
2. Say “density plot” (add “not a histogram” if needed).
3. Set x-axis limits 0 to 100, tick marks every 5 points, font size 14, and a color palette of your choice. Place the legend at the bottom.

Check: Is it a density curve (not bars)? Are ticks at every 5 from 0 to 100?

3) Add the Middle 95% to the Plot

Goal: Show the middle 95% region directly on the density plot.

1. Ask RTutor.ai to add vertical lines at the 2.5th and 97.5th percentiles and label both lines.
2. Ask to shade the region between the two lines. If shading fails, keep the labeled vertical lines only.

Check: Are both percentile lines placed and labeled correctly? If not, try “quantiles” instead of “percentiles.”

4) Interpretation & Extension

1. Write two sentences: describe the distribution shape (center, skewness) and state the middle 95% range and what it tells you about where most grades fall.
2. Repeat the full workflow (summary stats -> middle 95% -> density plot with lines) for Actual Exam Grade, then write the same two sentences for that variable.